

STATS 200A Lecture 3

Henry Liev

5 October 2026

Definition 3.1 (π -system)

Ω belongs, complements belong

$A, B \in \mathcal{P} \Rightarrow A \cap B \in \mathcal{P}$. Examples: Half-lines, half-open intervals together with $\emptyset$

Theorem 3.2 ($\pi - \lambda$ theorem)

$\mathcal{P}$ a π -system, $\mathcal{L}$ a λ -system, $\mathcal{P} \subseteq \mathcal{L} \Rightarrow \sigma(\mathcal{P}) \subseteq \mathcal{L}$

Proof. $\mathcal{D} := \lambda(\mathcal{P}) \subseteq \mathcal{L}$

$\mathcal{D}$ is a π -system $\Rightarrow \sigma(\mathcal{P}) \subseteq \lambda(\mathcal{P})$

For each $A \in \mathcal{D}$, the family $\mathcal{G}_A = \{B \subseteq \Omega : A \cap B \in \mathcal{D}\}$ is a λ -system.

$$A, B \in \mathcal{P} \Rightarrow A \cap B \in \mathcal{D}$$

$$A \in \mathcal{P}, B \in \mathcal{D} \Rightarrow A \cap B \in \mathcal{D}$$

$$A, B \in \mathcal{D} \Rightarrow A \cap B \in \mathcal{D}$$

■

Theorem 3.3

Let $\mathcal{P}$ be a π -system and $\mathbb{P}, \mathbb{Q}$ probability measures on $(\Omega, \sigma(\mathcal{P}))$. If $\mathbb{P} = \mathbb{Q}$ on $\mathcal{P}$, then $\mathbb{P} = \mathbb{Q}$ on $\sigma(\mathcal{P})$.

Proof. $\mathcal{L} = \{A \in \sigma(\mathcal{P}) : \mathbb{P}(A) = \mathbb{Q}(A)\}$ contains Ω , is closed under complements ($\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$) and disjoint countable unions (σ additivity). A λ -system containing $\mathcal{P}$; Dynkin gives $\sigma(\mathcal{P}) \subseteq \mathcal{L}$. ■

$$\lambda(\{2\}) = \lambda\left(\bigcap_{n \rightarrow \infty} \left(2 - \frac{1}{n} > 2\right)\right) = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

Definition 3.4 (Algebras)

$\mathcal{A} \ni \Omega$, closed under complements and *finite* unions.

Definition 3.5 (A *premeasure* on an algebra $\mathcal{A}$)

$\mu_0 : \mathcal{A} \rightarrow [0, \infty]$, $\mu_0(\emptyset) = 0$, countably additive *whenever the union stays in $\mathcal{A}$* .

$A_1, A_2, \dots$ disjoint and $\bigcup_{n \geq 1} A_n \in \mathcal{A} = \mu_0(\bigcup_{n \geq 1} A_n) = \sum_{n \geq 1} \mu_0(A_n)$

Theorem 3.6 (Caratheodory)

A premeasure on $\mathcal{A}$ extends to a measure on $\sigma(\mathcal{A})$. If σ -finite, uniquely

Definition 3.7 (Covering Cost)

$$\mu^*(\mathcal{S}) = \inf \left\{ \sum_n \mu_0(A_n) : A_n \in \mathcal{A}, \mathcal{S} \subseteq \bigcup_n A_n \right\}$$

Definition 3.8 (Measurable set)

E is measurable if for every test set $\mathcal{S} \subseteq \Omega$,

$$\mu^*(\mathcal{S}) = \mu^*(\mathcal{S} \cap E) + \mu^*(\mathcal{S} \setminus E).$$

Corollary 3.9

Lesbesgue measure λ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with $\lambda((a, b]) = b - a$ exists and is unique

Proof. *length is a premeasure*

If $(a, b] = \bigsqcup_n (a_n, b_n]$ then $b - a = \sum_n (b_n - a_n)$.

$\geq$ is a finite sub-collections;

$\leq$ uses compactness of $[a + \varepsilon, b]$ to reduce to finitely many intervals. ■

Definition 3.10 (Distribution function or CDF)

Let μ be a probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Its *distribution function* (or CDF):

$$F(x) = \mu((-\infty, x])$$

Theorem 3.11

(a) F is nondecreasing, right-continuous, $F(-\infty) = 0, F(+\infty) = 1$;

$$\mu(\{x\}) = F(x) - F(x-); \mu((a, b]) = F(b) - F(a)$$

(b) F determines μ

(c) Every F with the properties in (a) is the distribution function of a unique probability measure on $\mathbb{R}$

Proof.

(a) monotonicity and continuity of μ (μ finite!)

(b) half-lines are a π -system generating $\mathcal{B}(\mathbb{R})$; uniqueness theorem

(c) $\mu_0((a, b]) = F(b) - F(a)$ on the interval algebra; right-continuity makes it a premeasure; Caratheodory ■

$$X^{-1}(\mathcal{B}) = \{\omega : X(\omega) \in \mathcal{B}\} = \{X \in \mathcal{B}\}$$

Definition 3.12

$X : (\Omega, \mathcal{F}) \rightarrow (E, \mathcal{E})$ is *measurable* if $X^{-1}(B) = \{X \in B\} \in \mathcal{F}$ for every $B \in \mathcal{E}$.